THE AI CAPABILITY CONTINUUM: A THREE STEP FRAMEWORK FOR AI SYSTEM DEVELOPMENT

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ABSTRACT

The development of AI systems differs significantly from traditional software engineering methodologies. This paper introduces the AI Capability Continuum, a three-phase framework that conceptualizes the evolution of AI systems. We analyze the capability curve of AI compared to traditional software, highlighting key distinctions and challenges. Our framework divides AI development into the Foundation Zone, Acceleration Ramp, and Maturity Plateau, providing a structured approach to understanding AI progression. Furthermore, we discuss the impact of Large Language Models (LLMs) on AI development cycles and the strategic implications for organizations adopting AI technologies.

Keywords AI System Development · ROI in traditional software · Phases of AI Development

1 Introduction

Recent advancements in Artificial Intelligence (AI), fueled by open-source innovations and API-driven accessibility, have accelerated AI adoption across industries. This proliferation, supported by substantial funding, has led organizations—from startups to multinational corporations—to integrate AI into their products and services. However, despite this momentum, AI deployment remains fraught with challenges.

Empirical studies reveal that 96

Traditional software engineering has long relied on established methodologies—Waterfall, Agile, DevOps—to ensure efficiency and scalability. In contrast, AI software development (Software 2.0) introduces inherent complexities that necessitate a distinct framework. While existing models such as Microsoft's Team Data Science Process, CRISP-DM, and IBM's AI Ladder provide high-level guidance, they lack a structured mechanism for evaluating AI capability progression over time.

To address this gap, we introduce the AI Capability Continuum, a three-phase framework that conceptualizes AI system development through a capability curve. This curve, where system capability is plotted against time, provides insights into the evolution, maturity, and optimization of AI systems. Furthermore, we explore the impact of commoditized Large Language Models (LLMs) like ChatGPT, Claude, and Cohere on this continuum. LLMs enable rapid prototyping and MVP deployment, accelerating early market validation and reshaping the ROI curve for AI investments.

This framework offers a systematic approach to AI adoption, bridging the gap between theoretical advancements and practical implementation, ultimately guiding organizations toward sustainable AI success.

2 CAPABILITY CURVE OF TRADITIONAL SOFTWARE SYSTEM

Consider a traditional software system. If we plot a 2D curve with capability on the Y-axis and time on the X-axis, how does its growth evolve?

Initially, progress is slow due to extensive requirement gathering, scoping, budgeting, and team formation. This results in a flat slope. Once development begins, modern tools—agile methodologies, cloud computing, DevOps automation—drive a steep acceleration in capability.

However, as complexity and technical debt accumulate, further enhancements require disproportionately higher effort. The curve tapers off, reflecting diminishing returns. This trajectory is fundamental to understanding software system lifecycles (Fig. 1).

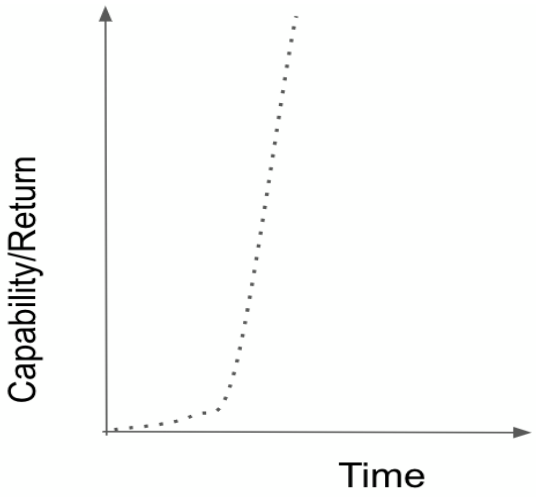

Figure 1: ROI/Capability evolution curve for traditional software systems.

3 CAPABILITY CURVE OF AI SYSTEM

When it comes to the capability curve of a typical AI project/system, any guesses what the shape might look like? it is very different:
Immediately, one sees distinct similarities and notable differences between this curve and the previous one. Unlike the sharp rise in Figure 1, the initial segment in Figure 2 remains flat for a much longer period, giving it a more elongated shape. After this slow start, the curve rises quickly, but not as steeply as in Figure 1. What is most striking is the sharp stagnation towards the end, where further improvements in capability slow down significantly.

The capability curve for AI often surprises industry practitioners, particularly those with a strong software engineering background. Founders, CXOs, and leaders typically expect AI systems to follow the same steep hockey-stick trajectory as traditional software systems. However, this is far from the reality.

The reason behind this difference lies in the fact that the well-established playbooks of traditional software development do not apply to AI development. This significantly affects the AI capability curve. To understand why the curve behaves this way, we divide it into three main segments: Foundation Zone (O-A), Acceleration Ramp (A-B), and Maturity Plateau (B-M). We'll examine each segment in detail to gain a deeper understanding of AI development and begin crafting its own playbook.

3.1 Foundation Zone (O-A)

1. Clarify the business problem – Develop a deeper understanding of the (often vague) business problem.
2. Define the problem – Transform (often vague) problem statement into a clear and well-defined business problem.
3. Evaluate AI suitability – Assess if the problem (step [2]) is even an AI problem or not
4. Set success metrics – Define both business and ML metrics for measuring success.
5. Data requirements – Understand the type and quantity of data needed to solve the AI problem.
   (a) Data availability – Verify if your organization has the kind & quantity of data needed.
   (b) Data preparation – Get the data needed, preprocess & clean it to create a v0 dataset.

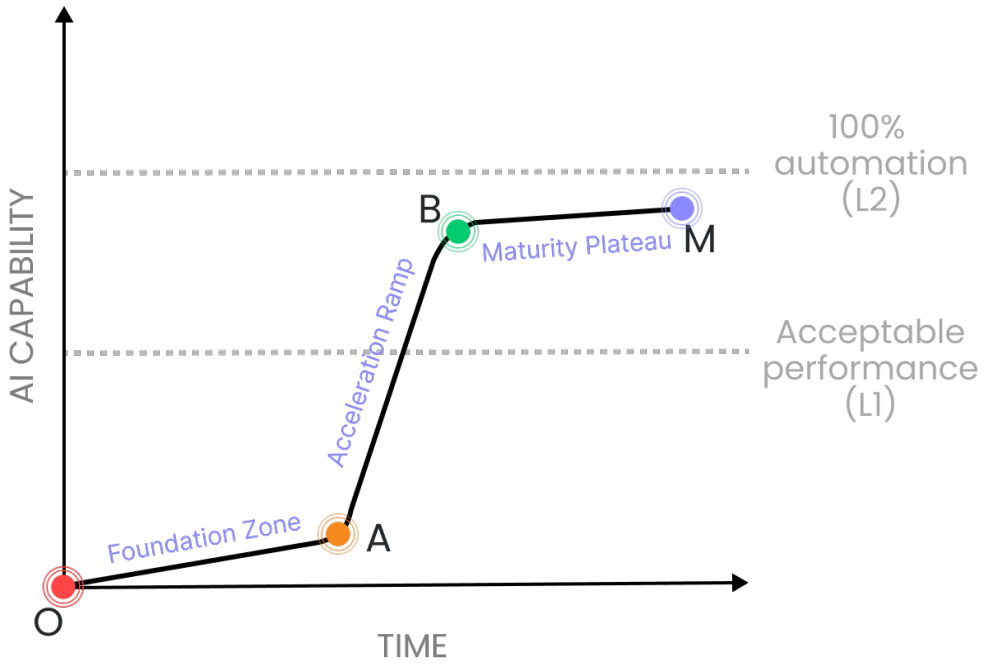

Figure 2: ROI/Capability curve for AI systems.

6. AI problem type – Determine the type of AI problem (e.g., supervised, unsupervised, reinforcement learning).
7. Understand the constraints your solution must adhere to:
   (a) Cost of mistake/wrong prediction
   (b) Prediction time - Real-time or batch?
   (c) Compute location - cloud or edge?
   (d) Any other constraints the solution must adhere to
      i. Time to prediction, explainability of prediction etc
8. Build a model – Develop a simple baseline model, or leverage third-party APIs.
9. Test business hypothesis – Quickly test the core business hypothesis - does your solution solve a major pain point for the end user (or business unit) or not? how likely is it to see PMF?
   (a) Market validation - If successful, how much is the market willing to pay for it?

The duration of the Foundation Zone (O-A) varies based on the problem's complexity, the organization's AI maturity, and the time required to test the core thesis. This phase can take anywhere from 30 days to 6 months. The most challenging aspects are typically dataset preparation and validating the core business hypothesis.

3.1.1 Observations

1. Many teams make the mistake of diving straight into building 'the' model, often aiming for state-of-the-art LLMs from day one. This is not the goal of the Foundation Phase. Instead, this phase is about answering fundamental questions:
   (a) Are we solving the right problem?
   (b) Is this even an AI problem, or can it be solved traditionally?
   (c) Do we have the right data to solve it effectively?
   (d) Most importantly: Is this a painkiller or a vitamin for the market? What is the market willing to pay?

   AI is expensive—data, compute, and talent don't come cheap. There's no point in building a state-of-the-art model if each prediction costs $3butthemarketwillonlypay1$ (or worse, has no interest at all).

   Also, building a great model is hard and takes time. It requires deep problem understanding, the right skill set, quality datasets, and heavy compute. Why invest at this level unless you're sure there's a viable business?

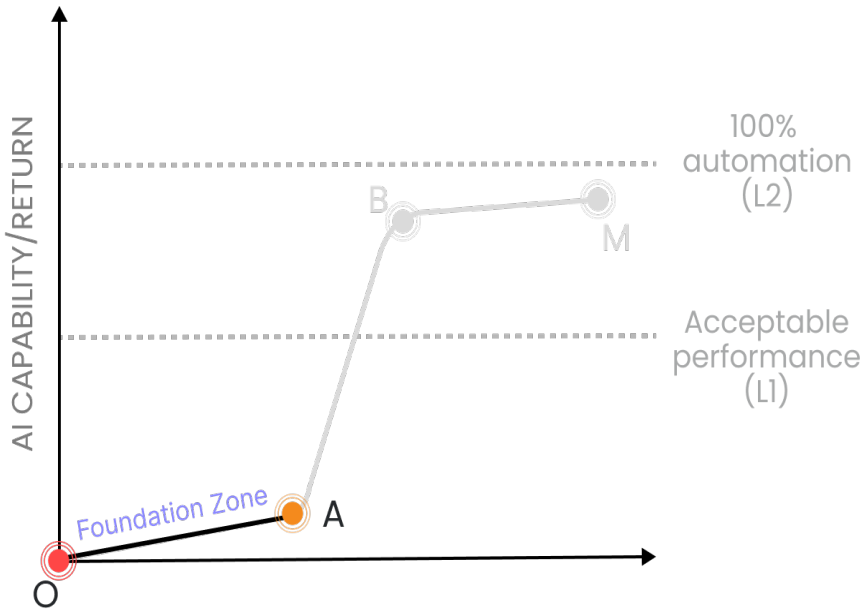

Fig 3: Foundation Zone Phase of capability curve for a typical AI system

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Seasoned entrepreneurs know that building a business is far harder than building a model. So why fixate on an advanced model from day one?

2. The goal of this phase is not to build the best model (unless you're a research lab). Instead, it's about tackling the hardest part of the problem: the business direction. Key questions to answer:
   (a) Painkiller or vitamin? Does this solve a major pain point?
   (b) Business viability: Does the AI system meet its business objectives? (e.g., Alexa was a great AI product but failed commercially)
   (c) Market willingness: Will users/clients pay for it? If so, how much?
   (d) Profitability: Can you extract enough LTV to offset high dev costs, indirect expenses, and margins to build a sustainable business?

   This is the core difference between AI in product companies vs. research labs. Academia pushes the state-of-the-art; businesses need viable, profitable AI solutions.

   To illustrate this, see the thought experiment on the next page.
3. In mature organizations, data is systematically organized for seamless access, much like a professional kitchen where chefs can quickly find ingredients. In contrast, data-immature organizations resemble a chaotic pantry—where retrieving specific data becomes a project in itself, delaying AI initiatives. This is a critical oversight among founders and executives. In fact, most AI projects in such organizations fail in the Foundation phase due to poor data strategy. If you're serious about AI, fixing your data strategy is non-negotiable.
4. Many teams, founders, and executives fixate on the model while overlooking a critical step—defining clear success metrics. AI systems are stochastic and error-prone, making it essential to establish both AI and business metrics from day one and ensure their alignment.

   Skipping this is like building a rocketship without a control panel—you won't know if you're on the right trajectory.
5. You may wonder—what about the model? We never discussed it, and that's intentional. Use whatever gets the job done—a third-party API, open-source model, or even humans making real-time predictions! As we emphasized earlier, this is not the phase to build a great model. The goal is to quickly assemble an AI system and test the market immediately.
6. If you're not training your own v0 model, you can deprioritize dataset creation and labeling initially. However, once your project moves beyond Point A, a quality dataset becomes essential for training models. It's best to start early to assess potential risks and avoid setbacks later.

This phase highlights why the traditional corporate mantra of securing “early wins” doesn’t apply to AI projects. The Foundation phase is inherently challenging, and quick wins are rare.

To summarize, the mantra for this phase should be “Quickly assemble & test the markets”

In a nutshell, this phase revolves around the mantra "Make it work". It closely resembles the pivotal 0-1 phase of a startup’s journey where the primary object is to get early feedback and figure out PMF.

3.1.2 Acceleration Ramp (A-B)

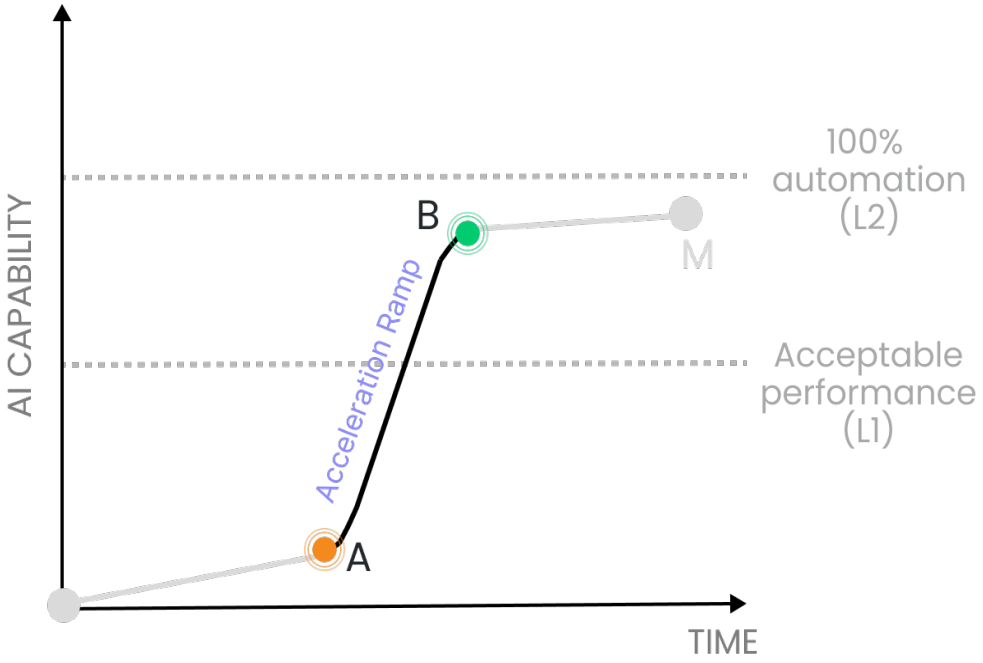

Figure 4: Acceleration Ramp (A-B) of capability curve for a typical AI system.

This is the middle part of the curve in Fig. 4 - Acceleration Ramp, where you experience the highest progress per unit effort.

At this stage, substantial work has already been done in the Foundation phase, including:

- Refining the problem statement
- Collecting data
- Operationalizing metrics & measurement methodology
- Gaining early validation from stakeholders (internal/external)

In this phase, the focus shifts to improving the model—moving from a model to the model. Key steps in this phase include:

1. Experimentation: Try various algorithms and build different models.
2. Benchmarking: Rigorously benchmark the performance of each model.
3. Threshold Achievement: Once the system crosses the acceptable performance level (L1), move the AI system to production and expose predictions to end users.
4. Continuous Improvement: Continuously push the best-performing model to production.
5. Testing & Replacement: Thoroughly test the new model and system, replacing the existing model with the new one.
6. Monitoring: Monitor the improvement in both ML metrics and business metrics.

Continue iterating through the previous steps, rapidly transitioning from one model to significantly better models. Over time, however, the gains will start to plateau, indicating that you've reached Point B. The duration of this phase varies depending on the specific problem and the AI maturity within the organization or team, typically lasting anywhere from 3 to 12 months.

3.1.3 Key Insights from the A-B Phase

Within the **A-B** Phase, several noteworthy observations come to light:

1. This phase begins when your product or offering's value proposition starts gaining traction in the market. The demos you carefully orchestrated to test the market have resonated well, and early customers—some even paying—start to come on board.

   With this traction, it becomes crucial to significantly ramp up the AI capabilities of your system. This is the ideal time to enter the Acceleration Ramp phase, where the focus shifts to strengthening the core: the model.
2. This phase is characterized by numerous iterations, as the goal is to transition from a model (simpler approaches) to the model (more complex approaches).
3. The reason for not jumping straight into the most advanced algorithm is rooted in Occam's Razor, a fundamental principle of AI: the aim is to find the simplest model that performs the best.
4. A key component of this phase is comparing multiple candidate models. The groundwork laid in the previous phase—focusing on metrics, measurement methodology, and rigorous test sets—serves as the foundation for this phase and proves invaluable. Without this preparation, it becomes challenging to critically evaluate and compare different models effectively.
5. At each step, you closely analyze the mistakes your model is making, and often, this analysis leads to the development of the next iteration of the model.
6. As you progress to better models, you continue deploying them to production. This is the stage where you evolve from the first version of your AI system to the third or fourth version.
7. Eventually, you'll notice that the gains begin to plateau, signaling that you're approaching Point B.

In short, this phase is all about "Make it Better."

3.1.4 Maturity Plateau (B-M)

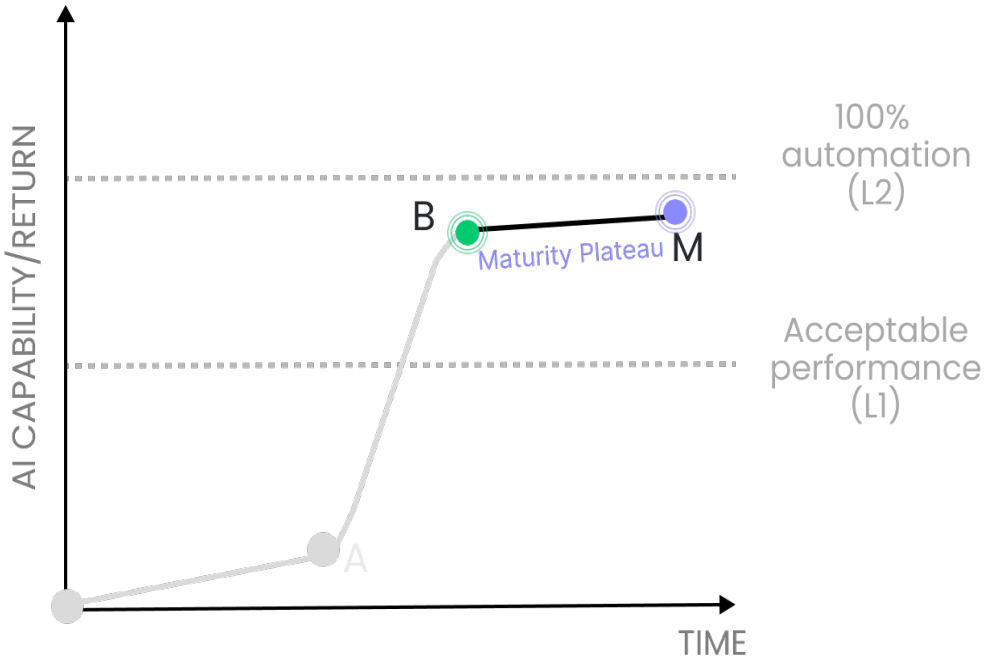

Figure 5: Fig 5: Maturity Plateau Phase of capability curve for a typical AI system

This is the final stretch. The curve starts to taper - your gains again start to move very slowly. This phase is about pushing the system to its limits. Your team:

1. Builds the 20th, 30th, or even 40th model iteration; 5th/7th/10th version of system.
2. Pushes system performance to its absolute limits.
3. Analyzes into every single edge case that your AI system is getting wrong.
4. Experiments with novel architectures/approaches/algorithms and, not just existing algorithms.
5. Tweaks core mathematical formulations of AI model and develops new mathematical formulations.

The **B-M** phase is marked by an unwavering commitment to reaching the utmost potential of the AI system. Its duration can range from 6 to 24 months, contingent upon the complexity of the problem at hand.

Despite massive effort, improvements are often just 1-2%. This phase can last 6 to 24 months, depending on complexity of the problem.

3.1.5 Key Insights from the B-M Phase

In the **B-M** phase, several notable observations come to the forefront:

1. This phase happens only when AI problem at hand is at the very core to your business, and you're scaling fast with strong traction and loyal, paying customers who not only love your product but also refer it to others (network effect)
2. Even a 1-2% AI improvement in the AI system will move your business numbers significantly. Say 5-10
3. Few AI projects reach this stage.
4. This phase is all about first-principles approach to model building—rethinking and rebuilding the very underlying math of the model from the ground up.
5. Notably, the capability curve beyond M never touches L2 (100% automation) - today's AI lacks AGI's self-learning ability to continuously improve.
6. No matter the use case or model quality, AI will always struggle with a long tail of edge cases. AI lives in this long tail, constantly encountering failures. This has huge implications for profit margins—but that's a topic for a different chapter.
7. AI lives in the long tail of edge cases, where progress slows due to rising complexity of handling edge cases. The more advanced the model, the greater its accuracy—demanding even higher-quality data for further improvements.
8. Instead of exponential performance gains, paradoxically one encounters an exponential rise in costs and effort. Example: Self-driving cars.

In summary, the **B-M Phase** is all about "Pushing the limits of your AI system" and must be embarked on only when AI aligns closely with the core business, thrives, and any gain in AI capability results in significant gains in business. Its duration can range from 12 to 36 months, contingent upon the complexity of the problem at hand and the utility of AI to business metrics.

In one line, this phase is about "Make it Great".

Paragraph Remarkably, of the three AI project phases you find yourself in, has significant implications for the type of AI talent you need for that particular phase. We'll delve into this topic in a separate article.

4 Expectations vs. Reality

Now that we have a comprehensive understanding of the AI Maturity Continuum, let us revisit our initial expectations regarding the ROI curve of a typical AI project:

Expectations:

- Take a business problem statement

- Build a state-of-the-art model, deploy it.
- Done

Now that we fully understand the AI Maturity Continuum, let's revisit our initial expectations for a typical AI project lifecycle. Our expectation:

Sounds simple, right? Reality is far from it.Fig. 6 sums this up:

Reality: A stark contrast to the above! Fig 6 effectively encapsulates the divergence between expectations and reality:

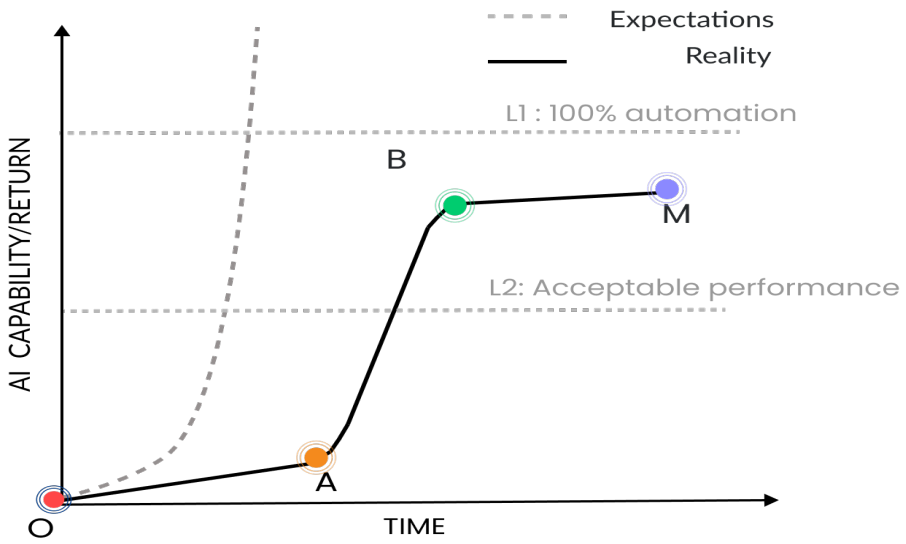

Fig 6: Capability curve for AI systems: Expectation vs Reality

Figure 6: Capability in AI: Expectations vs Reality.

This graphic visually depicts the significant disparity between what we might initially anticipate in an AI project and the multifaceted journey that unfolds in actuality.

5 Impact of ChatGPT/LLMs

Let's address the elephant in the room: How has the advent of LLMs like ChatGPT, Claude, Cohere, Lama, and others affected the AI Capability Continuum? The commoditization of these foundational models via APIs has provided a major boost to AI projects by offering high-level intelligence off the shelf, eliminating the need for extensive training. However, these models are not flawless and still make mistakes. While they accelerate early progress, refinement and adaptation are still required to tailor them to specific use cases.

This means that with off-the-shelf solutions like these, a savvy Product Manager (PM) paired with a driven Business Leader (BL) can quickly assemble a v0 solution without needing an AI team. Using prompt engineering and some help from a resourceful aspiring engineer, they can put together an AI system within 7-10 days, ready for a market demo. This aligns perfectly with the objective of the Foundation phase.

With the demo in hand, they can now aggressively market their "solution" and test the waters—are users excited about it? Are they willing to pay for it? How much are they willing to pay? This is one of the most critical aspects of the Foundation phase.

5.1 Key Insights

1. The O-A phase now is broken into 2 parts: O-X and X-A. O-X has all the steps of O-A barring the training set preparation step. This step is now part of X-A. Since dataset preparation step invariably takes time, X-A is still fairly flat.
2. O-X is steep but not as steep as A-B. This is because O-X still has a fair amount of fundamental steps (from O-A) to be executed. Also, good prompt engineering or fine-tuning LLMs for your use case takes some (if not a lot) of time. In contrast, As discussed earlier, one can generally move much faster in A-B.

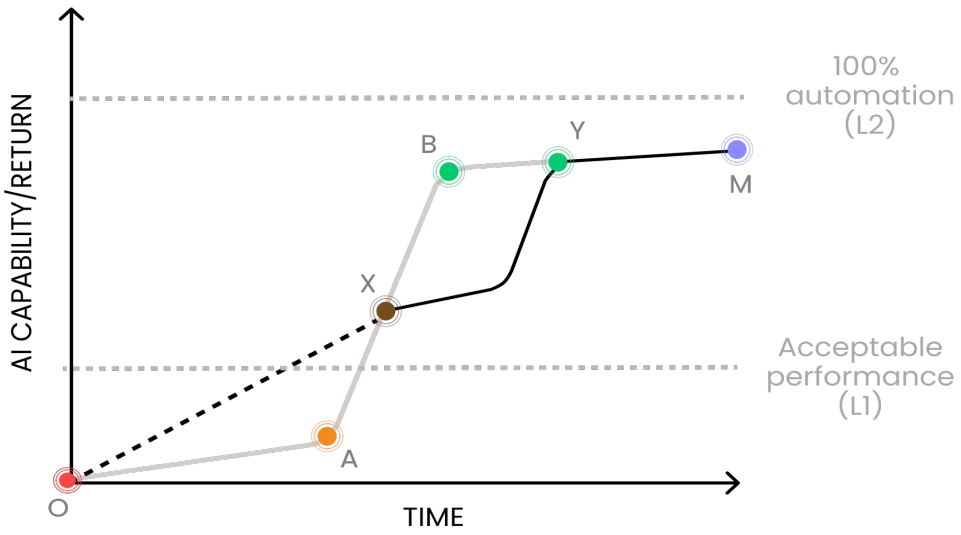

Fig 7: Impact of Off-the-shelve-LLMs on the capability curve

Figure 7: Impact of Chat-GPT/of-the-shelf LLMs on ROI/Capability in AI.

3. Since O-X uses state-of-the-art Chat-GPT/LLMs, one generally crosses acceptable [1] performance fairly quickly.
4. If the problem at hand is core to your business, you will have to train your own models. It can only be delayed, but can never be avoided. This journey remains the same, hence X-M.

Let's map this to the curve: Using chatGPT or off-the-shelf LLMs via APIs, you can quickly achieve a high level of intelligence, represented by point 'X' in Figure 7. The speed at which you reach this point creates a straight line from point 'O' (start) to point 'X'. In most cases, this level of intelligence is sufficient to start testing and selling your product or offerings in the market. At some point, as your paying customer base grows, their demands for model accuracy will increase. ChatGPT and similar off-the-shelf models, being generic systems trained on broad datasets, will only take you so far. Despite extensive prompt engineering, they reach a limit. To improve your system's performance, your next logical step is to fine-tune open-source foundational models using your own data. This will require you to gather data and build your own datasets. Once you've done that, the next step is to train foundational models from scratch. This journey mirrors the progression we saw in the Foundation Zone, Acceleration Ramp, and Maturity Plateau (O-A-B-M) in Figure 4. The only difference is that with higher accuracy and ML metric values, the curve beyond point 'X' will take a similar shape, but with greater complexity and refinement. It's important to highlight that AI is still a niche technology, and its landscape is rapidly evolving. In the future, due to the commoditization of AI, the cost, time and complexity required to develop AI systems will decrease significantly. As a result, the AI Capability Continuum curve could look very different. By using this curve, we can answer several critical questions: What is the right process for developing AI systems? From a talent acquisition standpoint, when is the right time to bring in AI scientists with PhD-level expertise? Why will AI companies' profit margins never exceed 40%, compared to SaaS companies, which enjoy 80-85% margins? We will explore these questions and more in the next set of articles.

6 Conclusion

The AI Capability Continuum provides a structured framework for AI system development, emphasizing the distinct phases of AI evolution: Foundation, Acceleration, and Maturity. By understanding the trajectory of AI capability growth, organizations can optimize investment strategies, align AI initiatives with business objectives, and set realistic expectations. While foundational models and LLMs have lowered entry barriers, achieving sustained AI success requires meticulous data management, iterative improvements, and domain-specific adaptation. Future research should focus on refining AI development methodologies, optimizing resource allocation, and addressing the long-term scalability of AI solutions.

[1] though what is 'acceptable' also keeps evolving

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